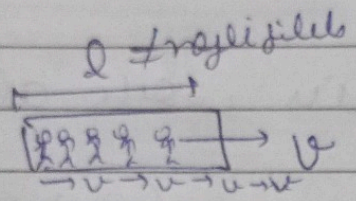
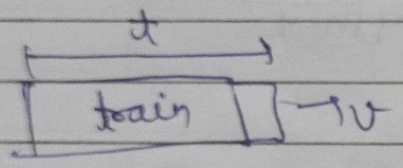
Time / Speed / Distance

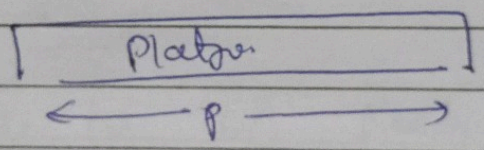
Km/hr → m/s × 5/18

m/s → km/hr × 18/5

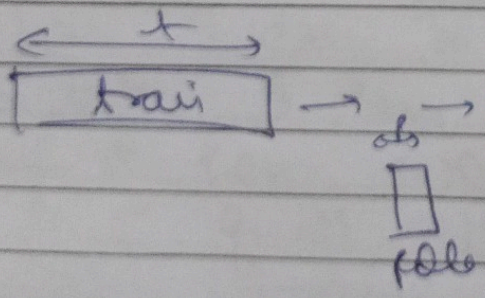
Train







(1) Train crosses a platform = P + t



(2) Train crosses a pole = t

$S = \frac{d}{t}$

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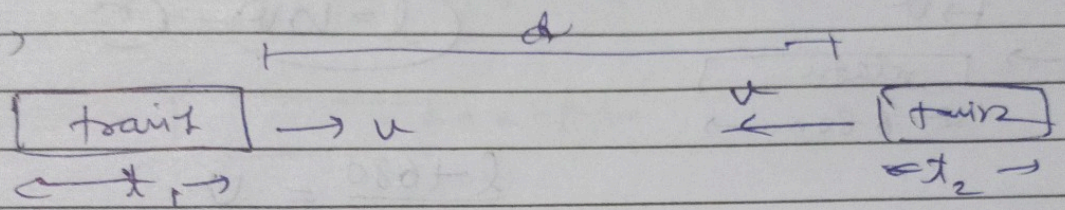
(3) Train crossing another lengthy body [another train, tunnel, bridge, platform] etc.:

distance covered = l + l_b
(length of train) (length of body)

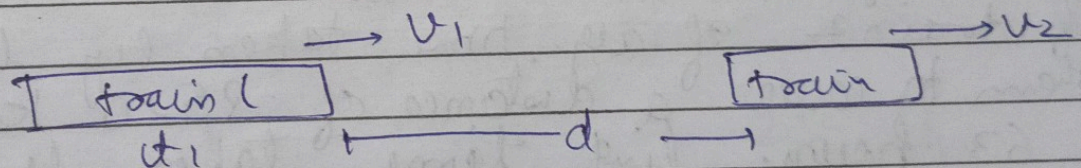
Case (i)



Relative speed = $v_1 + v_2$

Time of meet = $\frac{d}{v_1 + v_2}$

Case (ii)



Relative speed = $v_1 - v_2$ $(v_1 > v_2)$

time = $\frac{d}{v_1 - v_2}$

Q: A train crosses a pole in 10s and a 680m long platform in 30s. find length of train?

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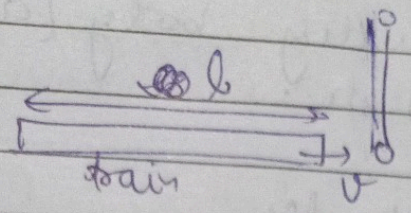
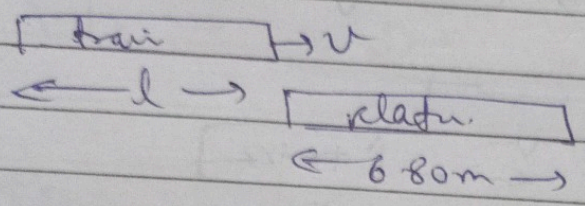


$\frac{D}{T} = S$

$\frac{l}{10} = v$

$l = 10v$ — (1)



$\frac{l+680}{30} = v$

$l + 680 = 30v$ — (2)

from (1) & (2)

$v = 34$

Q: The avg. speed of train 1 is 5 times than avg speed of train 2. If avg. time taken by both of them to cover a distance of 2520 km each is 63 hours. find time taken by train 1 to cover 1320 km.

Ans.

T1		T2
D_1	=	D_2
$V_1 \times t_1$	=	$V_2 \times t_2$
$5x \times t_1$	=	$x \times t_2$

$\frac{t_1}{t_2} = \frac{1}{5}$

$t_1 + t_2 = 63$

$t_1 = 21$

$t_2 = 42$

Train 1

$V = \frac{D}{T} = \frac{2520}{21} = 120$ km/hr.

$\frac{1320}{120} = 11$ hr.

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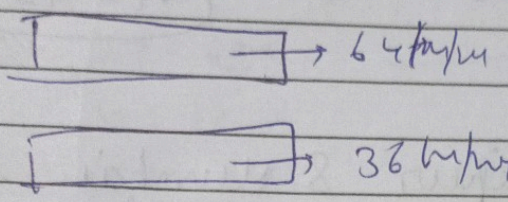
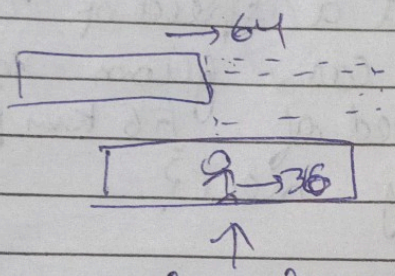
Q: Two trains are running on a parallel line in same direction at speed of 64 km/hr & 36 km/hr. respectively. Given that, faster train crosses a man in slower train in 36 sec, what is length of faster train?

Ans.





Speed = 64 − 36 = 28

l_f = length of faster train = distance covered

$d = v \times t = \left[28 \times \frac{5}{18}\right] m/s \times 36 \text{ seconds} = \underline{280 \text{ meters}}$.

Q: Train A can cross a pole & a platform length 450 m in 13s & 28s. Train B moving with speed of 81 km ph crosses Train A when moving in opp. dir^n in 16s. find length of Train B?

Ans.

pole		platform..
l	$\xrightarrow{+450}$	$l+450$.
13 s.	$\xrightarrow{+15 sec}$	28 sec.

$\frac{450}{15} \Rightarrow 30 m/s$.

$13 \times 30 \Rightarrow 390 m$

$l = 390 m$.

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	A	B
$l =$	390 m	x
$s =$	30 m/s	45/2 m/s.

$81 \times \frac{5}{18} = \frac{45}{2}$

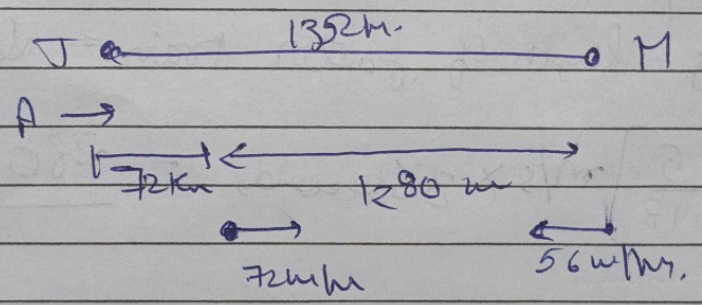
$$\frac{(390+x)}{105/2} = 16$$

$x + 390 = 840$

$x = 450$ m

Q: The dist b/w Jaipur & Mumbai is 1352 km. Amit starts driving his car from Jaipur to Mumbai at 9.40 AM at a speed of 72 km/hr. Mukesh starts driving his car from Mumbai to Jaipur at 10:40 AM at speed of 56 km ph. At what time will they meet?

A:



$$t = \frac{1280}{128 \text{ km/hr.}} \Rightarrow 10 \text{ hr}$$

8:40 PM

A

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Q: Train P is 450 m longer than train Q. When they moving in opposite direction, they cross each other in 68 sec. When they move in same dir^n, then the faster train crosses the slower train $4\frac{1}{4}$ min. If speed of faster train is 171 km/hr then find speed of slower train?

~~0.8 : 2.85~~ → $\frac{T_1}{T_2} \xrightarrow{\text{reciprocal}} \frac{S_1}{S_2}$

4 : 15

$\frac{19}{2} : \frac{11}{2}$

$\frac{68}{17 \times 15}$

19 : 11

171 : 99

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